Dr. Stefan Jackson, PhD (Marie Curie Fellow)

Scientist - Analytical | Formulation | Purification | Biologics | Small Molecules | Drug Delivery | Lyophilisation | Novartis BioCamp 2025 Winner | IVF 2026 Fully Funded Delegate

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PROFESSIONAL SUMMARY: Marie Skłodowska-Curie Fellow with a PhD in Chemical and Pharmaceutical Sciences & Biotechnology and 6+ years of R&D experience in analytical development, formulation, and characterization of biologics, small-molecule drug substances/products, and advanced drug delivery systems. Skilled in developing and applying state-of-the-art mass spectrometry methods, including LC-MS and GC-MS, to support process development, product characterization, and quality assessment against critical quality attributes (CQAs). Brings a collaborative, innovative, and patient-focused approach, working effectively across cross-functional teams to accelerate robust drug development.

KEY SKILLS & TECHNICAL EXPERTISE

Analytical Development, Mass Spectrometry & Characterisation: UHPLC, HPLC, LC-MS/MS (triple quad), HRMS (Q-TOF), GC-MS (HS, DI, SPME), ICP-MS, UV-Vis, NMR; MAM (multi-attribute method), impurity profiling, trace analysis, residual solvents, stability-indicating assays; DLS/zeta potential, TEM, AFM, SEM, FT-IR, DSC, rheology, optical microscopy; nanoparticle characterisation & synthesis (emulsion solvent evaporation, melt mixing, dissolution–dispersion); drug release kinetics modelling; MTT cytotoxicity; sample prep (SPE, LLE, SPME); method development, validation & transfer (ICH Q2).

Purification, DSP & Process Development: Preparative RP-HPLC, SEC, affinity chromatography, FPLC/ÄKTA (Unicorn); column packing/qualification; ultrafiltration/diafiltration, primary recovery, filtration, lyophilisation; high-throughput screening, scale-up, process optimisation; matrix-effect mitigation, recovery optimisation; technology transfer; small molecule & biologics (DS/DP) troubleshooting.

Quality, Regulatory & GMP Compliance: cGMP/GLP, ALCOA+ data integrity; CAPA, deviation/OOS investigations, root cause analysis (RCA); change control, SOP authoring; Quality Risk Management (FMEA, 5-Why, Ishikawa); CQA/QbD principles; ICH (Q2, Q3E, Q8–Q12), USP, Ph. Eur., FDA/EMA awareness; stability studies, validation & regulatory documentation; EHS/HSE standards.

Software, Data & Instrumentation: Agilent OpenLab/ChemStation, Waters Empower, MNova, TopSpin, OriginPro, Malvern Zetasizer, RheoCompass, UVProbe, SmartSEM, Spectrum 10; Microsoft Office; data analysis, trending & reporting.

Leadership, Collaboration & Core Strengths: Cross-functional collaboration (EU consortia); analytical & critical thinking; problem-solving; scientific/technical writing; stakeholder communication; project planning & execution; mentoring & leadership (PhD representative); adaptable, proactive, detail-oriented, results-driven, patient-centric mindset.

PROFESSIONAL EXPERIENCE

Marie Curie Research Scientist - Analytical & Formulation Chemistry
University of Camerino, Italy
February 2022 – December 2025

- Designed and formulated nanoparticle and in-situ gelling hydrogel drug delivery systems for sustained delivery applications; supported pre-formulation and optimization decisions.
- Led formulation studies involving crobenetine (BIII 890 CL) supplied by Boehringer Ingelheim, addressed low aqueous solubility via pH/excipient strategy and delivery-system design; developed and ICH Q2-validated an RP-HPLC-DAD/UV method for its release kinetics.
- Performed patch-clamp analysis via a collaborative work with Universidad Miguel Hernández de Elche on SH-SY5Y human neuroblastoma cells to quantify acute NaV1.2 inhibition.
- Acted as analytical SPOC across CMC/manufacturing; supported SOP/validation document review; ensured GMP/ALCOA+ compliance; coordinated MS-related activities and timelines.
- Presented scientific and technical analytical findings to multi-functional project teams, driving decisions in pre-formulation and formulation optimization.

Visiting Scientific Researcher – Polymer & Organic Chemistry (Industrial Experience)

Polypure AS, Oslo, Norway
May 2024 – July 2024
- Purified monodisperse PEG-lipid conjugates to more than 98% purity using FPLC, syringe and column chromatography; monitored purity by RP-HPLC-MS; prepared high-purity reference standards for downstream assay qualification.
- Performed SPE and LLE sample preparation for PEG-lipid/polymer matrices; optimised recovery and reduced matrix effects for trace-level HPLC and LC-MS assays.
- Synthesised PEG-lipid polyplexes for miRNA delivery; executed experiments in line with SOPs, ALCOA+, GMP and HSE requirements; supported OOS/OOX investigations and CAPA documentation.
- Coordinated lab workflows using digital project management tools, improving project efficiency and turnaround time.

Visiting Research Intern – Polymer & Nano Science
Mahatma Gandhi University, Kerala, India
February 2018 – May 2019
- Synthesised dextran/stearic acid/sesame oil hybrid nanoparticles for antibacterial edible coatings via emulsion solvent evaporation; characterised by DLS, zeta potential, FT-IR and TEM; biocompatibility confirmed by MTT assay and human fibroblast cell culture.
- Evaluated antibacterial activity against E. coli, P. aeruginosa and MRSA (disc diffusion); demonstrated antimicrobial efficacy and non-toxicity to human cells.
- Developed PLA/carbon black composites via two processing routes; performed ASTM-based mechanical (tensile, flexural, impact, HDT) and morphological (TEM, AFM, optical microscopy) characterisation; identified optimal filler loading for mechanical performance.

EDUCATION

Ph.D. in Chemical & Pharmaceutical Science and Biotechnology
University of Camerino, Italy (2022 – 2025)

M.Tech & B.Tech in Nanotechnology (Dual Degree)
Amity Institute of Nanotechnology, India (2014 – 2019)

AWARDS AND FELLOWSHIPS

- **Novartis BioCamp, 2025:** Winning Team for best AI driven pharmaceutical business pitch.
- **FameLab Camerino:** Secured first position in people's choice for the best 3 minutes science talk.
- **Marie Skłodowska Curie Fellowship:** European Union's Horizon 2020 Research and Innovation Programme. One of world's most prestigious and competitive fellowship.**'956477 grant number**.'
- **Elected PhD Representative**: School of Pharmacy, University of Camerino (2023 – 2025)

SELECTED PUBLICATIONS

- **MicroRNA-Based Delivery Systems for Chronic Neuropathic Pain Treatment in Dorsal Root Ganglion**; Pharmaceutics, 17(7), 930 (2025) doi:10.3390/pharmaceutics17070930.
- **Analysis of diffusion characteristics for aromatic solvents through carbon black filled natural rubber/butadiene rubber blends**; Polymer Composites, 42(1), 375-396 (2021) doi: 10.1002/pc.25832.

CERTIFICATIONS

Biotech Unveiled: Understanding the US biomedical innovation marketplace and its global role (by RA Capital Management)
Foundations of GMP (USAID/USP PQM+): Medicinal Products; API; Quality Risk Management; Deviations, Root Cause Analysis Tools and CAPA; Validation.
PIANO Network-Wide Training (MSCA ITN): Dorsal Root Ganglia techniques; Enabling Technologies; Nociception; Pain Management; Responsible Research & Innovation; Industrial Product/Process Development; Entrepreneurship & Innovation; Scientific Writing; Public Communication.
Occupational Health & Safety Training: Under D.Lgs. 81/08 - Generic (4h); Specific (8h) modules

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LANGUAGES: English - Fluent (CEFR C2; IELTS 8.0); **German -** Intermediate (CEFR B1; Certificate of German as Foreign Business Language); **Italian -** Working Proficiency; **Malayalam -** Native